ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

```java
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    System.out.println("enter a num:");
    int a=sc.nextInt();
    int b=sc.nextInt();

    System.out.println("sum:" +(a+b));
    System.out.println("differ:" +(a-b));
    System.out.println("product:" +(a*b));
    System.out.println("quotient:" +(a/b));

  }
}
```

6

OUTPUT

```
enter a num:
sum:11
differ:-1
product:30
quotient:0
```

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    System.out.println("enter a num");
    int year=sc.nextInt();

    if((year % 4==0 && year % 100!=0)|| year%400==0){
      System.out.println(year + " is a leap year");
    }else{
        System.out.println(year + " is not leap year");
    }
    }
  }

```

2000

OUTPUT

```
enter a num
2000 is a leap year
```

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    System.out.println("enetr a num");

    int n1=sc.nextInt();
    int n2=sc.nextInt();
    int n3=sc.nextInt();

    if(n1>=n2 && n1>=n3){
      System.out.println("max is:" +n1);
    }
    else if(n2>=n1 && n2>=n3){
      System.out.println("max is:" +n2);
    }
    else{
      System.out.println("max is:" +n3);
    }
  }
}
```

OUTPUT

```
enetr a num
max is:10
```

```
6
10
9
```

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

```
import java.util.Scanner;
public class fiboseries{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);

    System.out.println("Enter a  number");
    int maxnum=sc.nextInt();

    int firstterm=0;
    int secondterm=1;

    System.out.println("fibo series of " + maxnum + ":");

    while(firstterm<=maxnum){
      System.out.print(firstterm +" ");

      int nextterm=firstterm + secondterm;
      firstterm=secondterm;
      secondterm=nextterm;
    }
  }
```

5

OUTPUT

```
Enter a  number
fibo series of 5:
0 1 1 2 3 5
```

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

```
import java.util.Scanner;
public class fact{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);

    System.out.println("enter a pos no:");
    int num=sc.nextInt();

    long fact=1;

    for(int i=1;i<=num;i++){
      fact=fact*i;
    }
    System.out.println("factorial is:" +fact);
  }
}
```

5

OUTPUT

```
enter a pos no:
factorial is:120
```

RT-PATTERN

Write a Java program to print the following pattern using loops:*
**
***
****
*****

```
public class main{
  public static void main(String[] args){
    for(int i=1;i<=5;i++){
      for(int j=1;j<=i;j++){
        System.out.print("*");
      }
      System.out.println();
    }
  }
}
```

here!

OUTPUT

```
*
**
***
****
*****
```

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Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    int[] num={2,5,8,12,16,23,38,56,72,91};
    int target=23;

    int low=0;
    int high=num.length-1;
    int resultIndex=-1;

    while(low<=high){
      int mid=low+(high-low)/2;

      if(num[mid]==target){
        resultIndex=mid;
        break;
      }else if(num[mid]<target){
        low = mid+1;
      }else{
        high=mid-1;
      }
```

OUTPUT

```
element found at index:5
```

Your INPUT goes here!

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);

    System.out.println("enter a num:");
    int num1=sc.nextInt();

    System.out.println("enter a num:");
    int num2=sc.nextInt();

    int n1=num1;
    int n2=num2;

    while(n1!=n2){
      if(n1>n2){
        n1=n1-n2;
      }else{
        n2=n2-n1;
      }
    }
```

24
36

OUTPUT

```
enter a num:
enter a num:
The Gcd of 24 and 36 is: 12
```

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);

    System.out.println("Enter a num");
    int n=sc.nextInt();

    boolean isprime=true;

    if(n<=1){
      isprime=false;
    }else{
      for(int i=2;i<=n;i++){
        if(n%i==0){
          isprime=false;
          break;
        }
      }
    }
    if(isprime){
```

25

OUTPUT

```
Enter a num
25 is not a prime no
```

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    int[] arr={64,25,12,22,11};

    for(int i=0;i<arr.length-1;i++){
      int minIndex=i;
      for(int j=i+1;j<arr.length;j++){
        if(arr[j]<arr[minIndex]){
          minIndex=j;
        }
      }
      int temp=arr[minIndex];
      arr[minIndex]=arr[i];
      arr[i]=temp;
    }
    System.out.println("sorted array:");
    for(int num : arr){
      System.out.print(num + " ");
    }
  }
```

Your INPUT goes here!

OUTPUT

```
sorted array:
11 12 22 25 64
```

QUESTIONS

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run Save

```java
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc= new Scanner(System.in);

    System.out.println("enter a number");
    int n=sc.nextInt();

    for(int i=1;i<=5;i++){
      System.out.println(n + "X" + i + "="+(n*i));
    }
  }
}
```

OUTPUT

```
enter a number
5X1=5
5X2=10
5X3=15
5X4=20
5X5=25
```

INPUT

5

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QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

Run Save

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    long result=fact(n);
    System.out.println("fact of " + n + "is:" + result);
  }
  public static long fact(int n){
    if(n<=1){
      return 1;
    }
    return n*fact(n-1);
  }
}
```

OUTPUT

```
fact of 7is:5040
```

INPUT

7

LT-PATTERN

Write a Java program to print the following pattern using loops:*****
****
***
**
*

```
public class main{
  public static void main(String[] args){
    for(int i=5;i>=1;i--){
      for(int j=1;j<=i;j++){
        System.out.print("*");
      }
      System.out.println();
    }
  }
}
```

here!

OUTPUT

```
*****
****
***
**
*
```

PROGRAM EDITOR

Java

Run Save

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

```java
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int a=sc.nextInt();
    int b=sc.nextInt();

    int x=a;
    int y=b;

    while(y!=0){
      int temp=y;
      y=x%y;
      x=temp;
    }
    int gcd=x;
    int lcm=(a*b)/gcd;
    System.out.println(lcm);
  }
}
```

INPUT

```
18
12
```

OUTPUT

```
36
```

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int binary[]=new int[32];
    int i=0;
    while(n>0){
      binary[i]=n%2;
      n=n/2;
      i++;
    }
    System.out.println("Binary:");
    for(int j=i-1;j>=0;j--){
      System.out.print(binary[j]);
    }
  }
}
```

10

OUTPUT

```
Binary:
1010
```

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B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int binary=sc.nextInt();
    int decimal=0;
    int base=1;

    while(binary>0){
      int lastDigit=binary%10;
      decimal=decimal+lastDigit*base;
      base=base*2;
      binary=binary/10;

    }
    System.out.println("Decimal:"+decimal);
  }
}
```

1010

OUTPUT

```
Decimal:10
```

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    System.out.println("enter a num:");
    int num=sc.nextInt();
    int realnum=num;
    int sum=0;

    if(num<0){
      num=-num;
    }
    while(num>0){
      int lastdigit=num%10;
      sum=sum+lastdigit;
      num=num/10;
    }
    System.out.println("sum of digits" + realnum + "is:" + sum);
  }
}
```

2345

OUTPUT

```
enter a num:
sum of digits2345is:14
```

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int sum=0;
    if(n==0){
      System.out.println("product of digits: 0");
      return;
    }
    int product=1;
    while(n>0){
      int digit=n%10;
      product=product*digit;
      n=n/10;
    }
    System.out.println("product of digits: "+product);
  }
}
```

23

OUTPUT

```
product of digits: 6
```

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int base=sc.nextInt();
    int exp=sc.nextInt();
    int pow=1;

    for(int i=1;i<=exp;i++){
      pow=pow*base;
    }
    System.out.println(pow);
  }
}
```

2
3

OUTPUT

```
8
```

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    double radius=sc.nextDouble();
    double area=3.14*radius*radius;

    System.out.println(area);
  }
}
```

5

OUTPUT

```
78.5
```

Shpere

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    double radius=sc.nextDouble();

    double volume=(4.0/3.0)*3.14*radius*radius*radius;
    System.out.println(volume);
  }
}
```

6

OUTPUT

```
904.3199999999998
```

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    String str=sc.nextLine();
    String rev="";

    for(int i=str.length()-1;i>=0;i--){
      rev=rev+str.charAt(i);
    }
    if(str.equals(rev)){
    System.out.println("palindrome");
    }
    else{
      System.out.println("not palindrome");
    }

  }
}
```

OUTPUT

```
not palindrome
```

simats

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();
    int original=n;
    int sum=0;

    while(n>0){
      int digit=n%10;
      sum+=digit*digit*digit;
      n=n/10;
    }
    if(sum==original){
      System.out.println("Amstrong num");
    }
    else{
      System.out.println("not amstrong num");
    }
  }
}
```

OUTPUT

```
Amstrong num
```

153

RT-NUM.PATTERN

Write a Java program to print the following pattern using loops:1
22
333
4444
55555

```
public class main{
  public static void main(String[] args){
    Scanner sc=new Scanner(System.in);
    int n=sc.nextInt();

    for(int i=1;i<=n;i++){
      for(int j=1;j<=i;j++){
        System.out.print(i);
      }
      System.out.println();
    }
  }
}
```

OUTPUT

```
1
22
333
4444
55555
```

PROGRAM EDITOR

Java

Run Save

INPUT

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

```
import java.util.Scanner;
public class Perfect {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    int sum = 0;
    for (int i = 1; i < n; i++) {
      if (n % i == 0) {
        sum += i;
      }
    }
    if (sum == n) {
      System.out.println(n + " is a Perfect Number");
    }
    else
    {
      System.out.println(n + " is not a Perfect Number");
    }
  }
}
```

6

OUTPUT

```
6 is a Perfect Number
```

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Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
public class Main{
  public static void main(String[] args){
    int arr[]={25,36,44,52,65};
    int max=arr[0];
    for(int i=0;i<arr.length;i++){
      if(arr[i]>max){
        max=arr[i];
      }
    }
    System.out.println("max element is " +max);
  }
}
```

INPUT

Your INPUT goes here!

OUTPUT

```
max element is 65
```

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```java
public class Main{
  public static void main(String[] args){
    int arr[]={25,36,44,52,65};
    int largest=Integer.MIN_VALUE;
    int second=Integer.MIN_VALUE;
    for(int i=0;i<arr.length;i++){
      if(arr[i]>largest){
        second=largest;
        largest=arr[i];

      }
      else if(arr[i]>second && arr[i]!=largest){
        second=arr[i];
      }
    }
    System.out.println(second);
  }
}
```

OUTPUT

```
52
```

INPUT

15

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QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
public class Main{
  public static void main(String[] args){
    int arr[]={25,36,44,52,65};
    int min=arr[0];
    for(int i=0;i<arr.length;i++){
      if(arr[i]<min){
        min=arr[i];
      }
    }
    System.out.println("min element is " +min);
  }
}
```

OUTPUT

```
min element is 25
```

INPUT

Your INPUT goes here!

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
import java.util.Scanner;
public class main{
  public static void main(String[] args){
    int arr[]={12,23,32,34,56,65,76};
    int smallest=Integer.MAX_VALUE;
    int second=Integer.MAX_VALUE;

    for(int i=0;i<arr.length;i++){
      if(arr[i]<smallest){
        second=smallest;
        smallest=arr[i];
      }else if(arr[i]<second && arr[i]!=smallest){
        second=arr[i];
      }
    }
    System.out.println(second);
  }
}
```

OUTPUT

```
23
```

INPUT

Your INPUT goes here!

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

```java
public class main{
  public static void main(String[] args){
    int[]arr={23,34,45,56,58,65,76};

    for(int i=0;i<arr.length-1;i++){
      for(int j=0;j<arr.length-i-1;j++){
        if(arr[j]>arr[j+1]){
          int temp=arr[j];
          arr[j]=arr[j+1];
          arr[j+1]=temp;
        }
      }
    }
      for(int num:arr){
        System.out.print(num+" ");
    }
  }
}
```

Your INPUT goes here!

OUTPUT

```
23 34 45 56 58 65 76
```

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

```
import java.util.Arrays;
public class main{
  public static void insertionsort(int[] arr){
    for(int i=1;i<arr.length;i++){
      int key=arr[i];
      int j=i-1;
      while(j>=0 && arr[j]>key){
        arr[j+1]=arr[j];
        j=j-1;
      }
      arr[j+1]=key;
    }
  }
  public static void main(String[] args){
    int[] numbers = {12,11,13,5,6};
  System.out.println("original num: " + Arrays.toString(numbers));
  insertionsort(numbers);
  System.out.println("sorted array: " + Arrays.toString(numbers));
  }
}
```

OUTPUT

```
original num: [12, 11, 13, 5, 6]
sorted array: [5, 6, 11, 12, 13]
```

Your INPUT goes here!